



Oxidation of organophosphorus pesticides for the sensitive detection by a cholinesterase-based biosensor

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Abstract

A potentiometric flow injection-type biosensor developed in our laboratory was used for the determination of organophosphorus pesticides (OPs). The principle of the biosensor is that the degree of inhibition of a sensor enzyme by an OP is dependent on the concentration of the pesticide. The sensor system consisted of a reactor with acetylcholinesterase (AChE) immobilized on a controlled pore glass and a detector with a tubular H⁺-selective membrane electrode. In order to examine the possibility of enhancing the sensitivity of the sensor by converting OPs to oxidized forms (stronger inhibitors), a comparison of the degree of enzyme inhibition by OPs at 10⁻⁶ M before and after their oxidation was made. All of the ten pesticides tested exhibited greater inhibitory power toward the sensor enzyme following oxidation. All of the oxidized pesticides at 10⁻⁶ M inhibited the sensor enzyme to a considerable degree, demonstrating the utility of the developed method for the class-specific determination of OPs. A calibration curve for diazinon, over the concentration range of 10⁻¹¹–10⁻⁴ M, was obtained. The lower detection limit was 2 × 10⁻¹⁰ M. Treatment of the inhibited enzyme with pyridine-2-aldoxime restored the enzyme to near full activity, allowing repeated use of the sensor. © 2002 Published by Elsevier Science Ltd.

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1. Introduction

Organophosphorus pesticides (OPs) show low environmental persistence but display high acute toxicity. Their presence in water and food poses a potential hazard to human health and there is growing interest in their rapid and accurate determination. Conventional procedures, based on gas chromatography (GC) and high-performance liquid chromatography (HPLC), are reliable, but are expensive and require complicated and time-consuming sample pretreatment (Association of Official Analytical Chemists, 1995). Therefore, development of new methods for the rapid and inexpensive analysis of OPs, especially in field and on-line applica-

tions, is desirable. Biosensors seem to be suitable for such a purpose.

Recently, there have been numerous reports on the development of enzyme sensors based on the inhibition of cholinesterases (ChEs) for the detection of anti-ChEs, such as, OPs and carbamate pesticides (Skládal, 1996; Trojanowicz and Hitchman, 1996). The principle of these biosensors is based on the fact that the activity of ChEs is inhibited by these compounds and the degree of inhibition is dependent on the concentration of anti-ChE in contact with the enzyme. Acetylcholinesterase (AChE) and butyrylcholinesterase have been widely used as bioreceptors for these types of biosensors. According to the electrochemical method selected for the measurement of enzyme activity, a variety of potentiometric (Campanella et al., 1990; Tran-Minh et al., 1990; Kumaran and Tran-Minh, 1992; Campanella et al., 1996; Chung et al., 1998; Cho et al.,

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1999) and amperometric (Palleschi et al., 1992; Skládal, 1992; Skládal and Mascini, 1992; Bernabei et al., 1993; Marty et al., 1993; Cagnini et al., 1995; Martorell et al., 1997; Palchetti et al., 1997) biosensors were developed.

The flow injection system offers the advantages of highly reproducible timing and high sample throughput and eliminates the need to attain a steady state (Hansen, 1993). However, ChE-based biosensors developed to determine OPs were mostly of the probe-type and development of the flow injection-type has been rare (Kumaran and Tran-Minh, 1992). In our previous work, we developed a flow injection-type ChE sensor that consists of a tubular H^+ -selective membrane electrode and a reactor with AChE immobilized on controlled pore glass (CPG) (Chung et al., 1998). AChE catalyzes the hydrolysis of acetylcholine to yield choline and acetic acid. Thus, this potentiometric ChE-based biosensor can determine ChE activity by measuring the pH change that takes place with a H^+ -selective electrode. In this paper, we describe the application of this sensor to the analysis of ten OPs to examine the general applicability of the sensor to OP analysis.

There have been several reports that OPs show stronger inhibitory action when their thiophosphoryl group, $(RO)_2P(=S)-$, is oxidized to a phosphoryl group, $(RO)_2P(=O)-$ (Fallscheer and Cook, 1956; Chambers and Chambers, 1991). Therefore it was thought that conversion of OPs to the oxidized form (oxons), prior to analysis, would improve the sensitivity of ChE-based sensor. Kumaran and Tran-Minh (1992) utilized this observation to improve the sensitivity of their AChE sensor, which they developed for the determination of OPs. They used bromine as the in situ oxidant for the aqueous sample solutions and injected the oxidized sample solutions directly into the FIA system. In their study, the efficiency of the oxidation reactions was unknown and the possibility of enzyme inhibition by certain undesirable products, including the bromine species, was not addressed. Thus, it is uncertain whether or not the enhanced sensitivity, observed in their study, resulted purely from oxidation. In this paper, we describe the use of our ChE-based sensor for the comparison of the inhibitory power of ten OPs with their corresponding oxons. To examine the possibility of repeated use of the proposed sensor by reactivation of the inhibited sensor enzyme, the degree of reactivation of the inhibited enzyme by pyridine-2-aldoxime (2-PAM) was also studied.

2. Materials and methods

2.1. Chemicals and instruments

Polyvinyl chloride (PVC, high molecular weight), tridodecylamine and tetrahydrofuran (THF) were purchased from Fluka (Buchs, Switzerland) and dioctyl

phthalate from Aldrich (Milwaukee, USA). Sodium tetraphenylborate, acetylcholinesterase (EC 3.1.1.7, V-S type from electric eel, 1000–2000 units/mg protein), acetylcholine chloride, glutaraldehyde (25% aqueous solution), pyridine-2-aldoxime methiodide, HEPES buffer and aminopropyl CPG (200–400 mesh, average pore size of 500 Å, average amine content of 70 $\mu\text{mol/g}$) were obtained from Sigma (St. Louis, USA). The 10 OPs used (parathion-ethyl, parathion-methyl, fenthion, fenitrothion, diazinon, EPN, chlorpyrifos, dichlofenthion, bromophos-ethyl, and bromophos-methyl) were purchased from Dr. Ehrenstorfer GmbH (Augsburg, Germany). Bromine was obtained from Junsei Chemical (Tokyo, Japan). Analytical TLC (silica gel F₂₅₄, fluorescent) and preparative TLC plate (silica gel 60, fluorescent, 1 mm) were from Merck. All other chemicals that were used were of reagent grade quality or higher.

2.2. FIA manifold

The configuration of the FIA manifold is shown in Fig. 1 (Chung et al., 1998). The carrier stream was pumped through the FIA manifold by a multi-channel peristaltic pump (Ismatec model IPC). Samples were injected into the carrier stream using a four-way rotary injection valve (Rheodyne). The decrease of pH in the sample plug was sensed with a homemade membrane electrode and a double junction reference electrode (Orion 90-02), in combination with a pH/mV meter (Mettler model 340), and was recorded on a flatbed recorder (Kipp & Zonen model BD 111). The carrier consisted of a HEPES buffer, pH 7.2, containing 20 mM $MgCl_2$ and 100 mM NaCl. This buffer solution will be referred to as “HEPES working buffer” in the text. Reactors were assembled by packing AChE-immobilized CPG in a 2.06 mm i.d. Tygon tubing. The components of the system were connected with 0.89 mm i.d. Tygon tubing.

2.3. Construction of the detector, H^+ -selective membrane electrode

Construction of the detector, a tubular H^+ -selective membrane electrode (Meyerhoff and Kovach, 1983; Chung et al., 1998), was as follows. A casting solution for the H^+ -selective membrane was prepared by dissolving 20 mg of tridodecylamine, 132 mg of dioctyl phthalate, 1.4 mg of sodium tetraphenylborate, and 51 mg of PVC in 1 ml of THF. An 18 gauge syringe needle was inserted into 50 mm Tygon tubing (i.d. 0.64 mm) and a 5 mm hole was cut out lengthwise with a razor blade. Onto the cut-out region were poured 6 drops of the casting solution with the THF evaporated after each addition. After pulling out the needle, the membrane-attached tubing was inserted through the two holes drilled into the wall of a plastic centrifuge tube of 1.5 ml capacity (outer jacket) and the holes were sealed with a

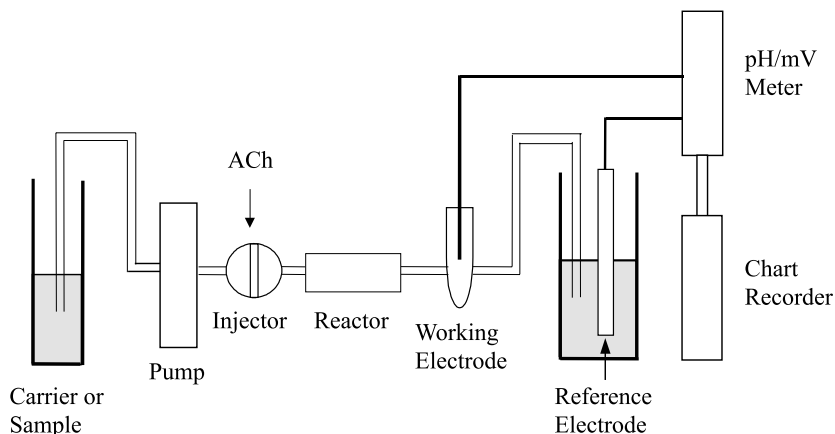


Fig. 1. Schematic diagram of AChE-based flow injection biosensor.

cyclohexanone-PVC paste (1 ml/1 mg). The detector was completed by pouring a 0.5 M citrate buffer solution (pH 4.5) and two drops of 10^{-2} M KCl into the jacket and placing an electroplated silver wire into the internal solution.

2.4. Immobilization of AChE

AChE was immobilized on CPG (Chung et al., 1998) as follows. To 1 g of aminopropyl CPG was added 3.3 ml of 2.5 % (m/v) solution of glutaraldehyde in 0.05 M phosphate buffer, pH 7 (PB). The mixture was shaken in a shaker for 1 h and then the solid washed with distilled water. To activated glass was added AChE (330 units) in 2.4 ml of PB, and the mixture was stored in a refrigerator at 4°C overnight. After being washed sequentially with distilled water, 1 M KCl, and 2.5 mM HEPES buffer (pH 7.5), the beads were stored in a refrigerator at 4°C until they were used.

3. Oxidation of OPs

Oxidation of OPs were accomplished by combining a pesticide solution (1 mM) in acetonitrile with an equal volume of bromine solution (10 mM) in acetonitrile and then vortexing for a few seconds. The products were isolated by preparative TLC and were characterized by ^1H NMR and GC/MS. The detailed procedures will be published elsewhere (Kim et al., 2001).

4. Measurement of the degree of enzyme inhibition

In order to compare the inhibitory power of unoxidized and oxidized OPs toward the sensor enzyme, the degree of inhibition of the sensor enzyme caused by unoxidized and oxidized pesticides at 10^{-6} M were de-

termined. The pesticide working solutions were prepared by dilution of the 5×10^{-2} M stock solution of the pesticides in ethanol with 2.5 mM HEPES working buffer, pH 7.2. The final working solutions contained ethanol in 2% to improve the solubility of the pesticides. Inhibition of the enzyme was determined by comparing the enzyme activity before and after passage of a pesticide solution through the reactor for a given period of time. The operating conditions of the sensor for these experiments were: CPG, 200–400 mesh (average pore size, 500 Å; amine content, 50–125 $\mu\text{mol/g}$); reactor length, 4 cm; carrier, 2.5 mM HEPES working buffer (pH 7.2); flow rate, 0.7 ml/min; substrate sample, 100 μl of 5×10^{-3} M ACh; pesticide sample, 10^{-6} M in 2% ethanol-HEPES. The procedure for measuring the inhibition of the sensor enzyme was as follows. The carrier buffer was passed through the sensor system until the base line was stabilized and then 100 μl of 5×10^{-3} M ACh in HEPES buffer was injected into the carrier stream, and the response recorded. Then the detector was disconnected from the system and the pesticide working solution was passed through the detector-free system for 30 min to inhibit the enzyme. The detector was then reconnected to the system and the carrier was passed through the system for 30 min. A solution of 100 μl of 5×10^{-3} M ACh in HEPES buffer was injected into the carrier stream as before and the response recorded. For diazinon and parathion-ethyl, percent inhibition of the enzyme by the oxidized diazinon over the concentration range of 10^{-11} – 10^{-4} M was determined in order to obtain a calibration curve.

5. Reactivation of the inhibited enzyme

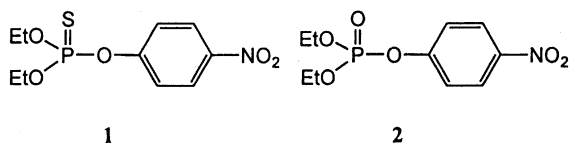
The ability of 2-PAM to reactivate the enzyme, inhibited by 10^{-6} M OPs, was measured. The enzyme was inhibited as described above and then treated with

20 μM 2-PAM in a HEPES working buffer for 20 min, followed by the passage of the HEPES working buffer for 20 min. Following these treatments, enzyme recovery was measured.

6. Results and discussion

6.1. Inhibition of the sensor enzyme by OPs

The degree of inhibition (percent inhibition) of the sensor enzyme by 10^{-6} M unoxidized and oxidized OPs, after 30 min of enzyme-pesticide contact time, are shown in Table 1. The typical response of the sensor, observed in the case of diazinon, is presented in Fig. 2. The oxidized OPs used were those obtained by a reaction with bromine. They were oxons resulting from oxidative desulfuration. For example, oxidation of parathion (1) yielded paraoxon (2).



In the case of fenthion (3), oxidation yielded two major products, i.e., the oxon and a byproduct that resulted from oxidation of the sulfide group to the sulfoxide (4) (Kim et al., 2001).

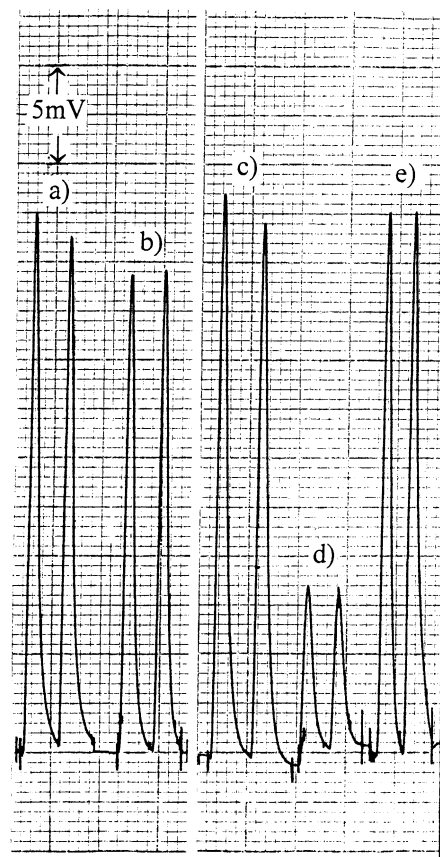
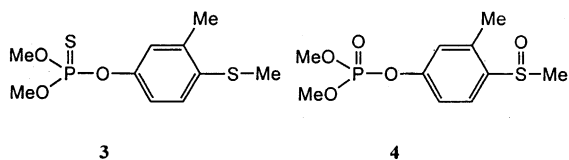


Fig. 2. Dynamic sensor responses (a) before enzyme inhibition, (b) after enzyme inhibition at 30 min incubation with 10^{-6} M diazinon, (c) after enzyme reactivation at 20 min incubation with 20 μM PAM, (d) after enzyme inhibition at 30 min incubation with 10^{-6} M oxidized diazinon, and (e) after enzyme reactivation at 20 min incubation with 20 μM PAM.

Table 1

Comparison of the percent inhibition of the sensor enzyme by organophosphorus pesticides before and after oxidation

Pesticides	Before oxidation		After oxidation	
	% Inhibition	% Reactivation	% Inhibition	% Reactivation
Parathion	17	100	77	97
Fenitrothion	7.7	104	31	89
Fenthion ^a	8.5	107	19	83
Fenthion ^b	7.7	103	37	70
Chlorpyrifos	3.9	100	69	80
Diazinon	13	107	73	95
EPN	5.6	100	80	109
Dichlofenthion	11	100	75	80
Parathion-methyl	11	100	48	93
Bromophos-ethyl	7.4	93	12	92
Bromophos-methyl	8.7	92	20	88

^a The product resulted from the oxidation of only the thione group.

^b The product resulted from the oxidation of both the thione and sulfide groups.

Since both products exhibited greater enzyme inhibitory power than the unoxidized pesticide, the formation of multiple products would not be a problem in the sensitive determination of fenthion.

The percent inhibition through use of oxidized OPs increased to between 1.6- and 18-fold, and the average increase was 3.5-fold. Therefore, it may be safely concluded that, due to the greater inhibitory power of OPs after oxidation, oxidation of OPs by bromine may be regarded as an excellent method for improving the sensitivity of the ChE-based sensor for the determination of OPs.

The plot of percent inhibition of the enzyme vs. concentration of oxidized diazinon over 10^{-4} M– 10^{-11} M is shown in Fig. 3. While the lower detection limit (10% inhibition) before oxidation of the pesticide was ca. 10^{-7} M, the limit after oxidation was 2×10^{-10} M, which corresponds to a 500-fold increase in sensitivity. The lower detection limit of diazinon with the present biosensor (2×10^{-10} M) is lower than that observed in a previous study (2×10^{-9} M) regarding the determination of diazinon with an amperometric ChE-based biosensor (Cagnini et al., 1995). Inhibition data at lower concentrations were also obtained for parathion-ethyl. The inhibition curve was similar but the detection limit was higher (10^{-9} M).

The percent reactivation of the inhibited enzyme at 20 μ M 2-PAM after 1 h incubation varied between 76% and 102% (Table 1). Reactivation, in some instances, was far from complete, however, it was generally observed that increasing incubation time eventually led to a full restoration of the enzyme activity. Feasibility of the repeated use of the sensor enzyme through reactivation

of the inhibited enzyme was confirmed in a series of measurements of enzyme inhibition at various concentrations of diazinon. All the measurements were made using a single reactor; measurement was begun at the lowest concentration (10^{-11} M) and then, the reactor was reactivated for the next measurement (10^{-10} M). This cycle was repeated until the final measurement, for the highest concentration, was made. Therefore, enzyme reactivation by 2-PAM allowed repeated use of the sensor enzyme for quite a large number of measurements.

7. Conclusion

The ChE-based flow injection biosensor, consisting of a reactor with AChE immobilized on CPG and a detector with tubular H^+ -selective membrane electrode, is adequate for the rapid analysis of OPs. Since OP pesticides show a greater inhibitory action toward AChE when oxidized, oxidation of OP pesticides may be considered a useful method to improve the sensitivity of ChE-based biosensors. Since the detection limit of this sensor for diazinon was 2×10^{-10} M, the sensitivity of the sensor is satisfactory. Treatment of the inhibited enzyme with 2-PAM restored the enzyme activity almost fully allowing repeated use of this sensor for OP analysis.

Acknowledgements

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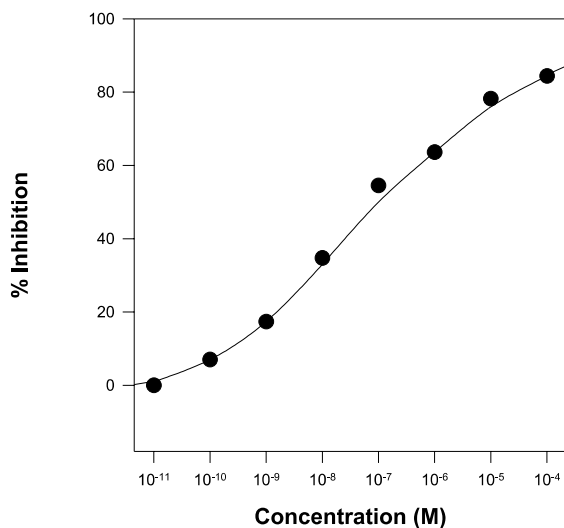


Fig. 3. Plot of percent inhibition of the sensor enzyme vs. concentration of oxidized diazinon.

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